Arjun Maneesh Agarwal

arjuna.ug2024@cmi.ac.in | thearjunagarwal.github.io | github.com/TheArjunAgarwal | LinkedIn

RESEARCH INTERESTS

Game Theory, Social Choice Theory, Algorithms, Probability Theory, and Functional Programming (Haskell). Generally, I am interested in using techniques from mathematics and computer science to model, predict and understand social and behavioral dynamics, and to use that understanding to design better mechanisms and make better choices.

EDUCATION

Chennai Mathematical Institute, Chennai, India — 2024 – 2027 (expected)
BSc (Hons) in Mathematics and Computer Science

- Recipient of the Shriram Scholarship (full tuition waiver and stipend).

RESEARCH EXPERIENCE

Committee Selection and Participatory Budgeting *Prof. Sanjukta Roy* — 2026 – present

- Working on *Nash Committee Selection* and *Participatory Budgeting*; manuscript in preparation.

Advanced and Recent Developments in Algorithms *Prof. Neeldhara Misra* — Fall 2026

- Reading recent papers in algorithms and working on open problems in graph algorithms, fair division and combinatorial game theory.

Summer Research Internship Programme, IIT Gandhinagar *Prof. Jyothi Krishnan* — Summer 2026

- Retraced and wrote an exposition of the cryptanalysis of the Enigma machine.
- Worked on solving, and proving the computational complexity of, Walter Joris' games *Sequentium* and *Blackhole*.
- Unpublished work: an Index-of-Coincidence-based method that breaks 300–500 character Enigma ciphertexts in under 10 seconds with high probability.

Voting Theory *Prof. Sanjukta Roy* — Summer 2025

- Read and presented chapters from the *Handbook of Computational Social Choice* (Ch. 2–7) and *Trends in Computational Social Choice* (Ch. 1–4).
- Seminar: *Manipulation, Control, Bribery and Barriers: Why rigging an election is too hard?*

Matching Theory *Profs. Prajakta Nimbhorkar and S. P. Suresh* — Spring 2025

- Implemented the Gale–Shapley and Irving algorithms in Haskell.
- Designed and ran a questionnaire-based matching event (*Friendship Finder*) that involved 58% of the campus; handled the algorithms, data security, and post-event data analysis. Code: github.com/TheArjunAgarwal/open-pact

Introductory Game Theory *Profs. Rajeeva Karandikar and Nagarajan Krishnamurthy* — Fall 2024

- Studied *Game Theory* (Maschler–Solan–Zamir), *Games of Strategy* (Dixit–Skeath–McAdams) and *Strategy* (Watson).
- Seminar on game-theoretic models of bubbles and bank runs, based on Moinas and Pouget (2014) and Golman, Jain and Saraf (2021).

PUBLICATIONS AND WRITING

Haskell for CMI (textbook, in progress) *with Ryan Hota and Shubh Sharma* — 2025 – present

- Used as a reference in the 2026 offering of Prof. S. P. Suresh's course on Haskell. haskell-for-cmi.vercel.app

TALKS AND SEMINARS

Fighting Uncertainty with Randomness: Online Algorithms *CMI Student Seminar, Aug 2026* — Slides

To Count Magic Squares, Talk to a Physicist *CUPIT Initiative, Sep 2026* — Video Slides

How to Crack Enigma? *SRIP 2026, Jul 2026* — Slides

Collecting Coins on Trees *Sprouts 2026, Apr 2026* — Slides Interactive

Friendship Finder: Algorithms, Data Analytics and Marketing *CMI Student Seminar, STEMS 2026* Slides Code Video

Manipulation, Control, Bribery and Barriers *CMI Student Seminar, Oct 2025* Slides

Rational Roots of Irrational Markets *CMI Student Seminar; STEMS 2025* Video

TEACHING

Teaching Assistant, Graduate Algorithms (CMI) *Prof. Samir Datta* Fall 2026

- Designed slides, handouts and problem sets; ran tutorials and took quizzes. thearjunagarwal.github.io/gralgo-tutorials

Teaching Assistant, Design and Analysis of Algorithms (NPTEL) *Prof. Madhavan Mukund* Fall 2026

Teaching Assistant, Computational Physics (CMI) *Profs. Ajit Kumar Mehta and Prashant Kocherlakota* Fall 2026

Teaching Assistant, Algorithm Design Techniques (MSc Data Science, CMI) *Prof. G. Philip* Spring 2026

- Designed slides, handouts, extra problems and interactives; ran tutorials and an ICPC-style contest. thearjunagarwal.github.io/ds-algo-tutorials

Teaching Assistant, Introduction to Programming with Haskell (CMI and NPTEL) *Prof. S. P. Suresh* Spring 2025, Fall 2025, Spring 2026

- Designed problem sets and assignments, ran tutorials, and co-authored the course textbook.

Teaching Assistant, Empower with English (CMI) *Prof. Usha Mahadevan* Fall 2025

Mathematical Olympiad Instructor *High school* 2022 – 2024

- One of my students went on to represent Nepal at the IMO.

SERVICE AND LEADERSHIP

Founding Editor and Writer, CMI Student Blog *Chennai Mathematical Institute* 2026 – present

- Co-authored the proposal, co-designed the site, and helped set editorial freedoms and the peer-review process. Launched July 2026.

Head Coordinator, Tessellate *CMI's annual fest* 2026

- Led sponsorship, outreach, planning and execution of CMI's annual cultural and technical fest.

Problem Setter and Physics Head, STEMS *CMI* 2025, 2026

- Set problems in Mathematics, Computer Science, Physics and Probability–Statistics.

Co-organizer, Integration Bee *CMI* 2025, 2026

- Built the website, question set, slides and format with Piyush Kumar Jha. thearjunagarwal.github.io/integrationbee

HONORS AND AWARDS

- **SMMC Merit Prize** (2025), for creative and insightful work on a problem; shared with Piyush Kumar Jha.
- National Olympiads (2019 – 2023): Mathematics (2019 – 2023), Informatics (2023), Astronomy (2022 – 2023), Biology (2022 – 2023), Junior Science (2020).
- Top 0.5% in NEET, IIT-JEE and ISI entrance examinations (2024).

SKILLS

- **Programming:** Haskell, Python, C++, R, Java, Rust, HTML, CSS, JavaScript.
- **Tools:** LaTeX, Typst, Markdown, Git.